AI Cockpit User Guide

The overlay, the full-screen chat app, actions, connectors, building a clean release, and extending it with new connectors and CLIs.

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1. Quick start

1. Start **AiCockpit.exe** from `dist\AiCockpit\` (or the desktop shortcut made by `install-shortcut.vbs`). It lives in the system tray.
2. Press **Ctrl+Shift+Space** (default hotkey) or click the tray icon to open the overlay. **Esc** hides it.
3. Type a question and press **Enter**. The answer appears in the Actions tab; the job appears in the Jobs tab.
4. Click **Full view** (or the expand icon on a job) to continue in the full-screen chat app.

> Billing: jobs run on your subscriptions (Claude Pro through Claude Code, Google AI Pro through the Antigravity CLI). API-key variables such as `ANTHROPIC_API_KEY` are removed from every job so it never bills per token.

2. The overlay

Part	What it does
Header	Reactor orb (idle / active / waiting for approval / error), subscription pills with usage status, routing policy pill, theme button, hide.
Prompt box	Enter runs it. Start with `/` to search actions. Prefix `/claude` or `/agy` to force a CLI. Ctrl+V pastes files or a screenshot.
Attach	Adds files, folders or images. Jobs can read them, never change them.
Save to	Opens a Windows folder picker. Every job saves into that folder until you clear it (× on the "Saving to" line). Also in Full view.
Continue chip	Shown after a finished chat. Selected: your next prompt is a follow-up in the same session. × starts fresh.
Terminal checkbox	Runs the job in an interactive Claude Code terminal instead of headless.
Kill all	Ends every running job immediately (grants revoked, processes terminated).
Tabs	**Actions** (conversation + action tiles), **Jobs**, **Activity** (every tool call, audited), **Connectors**.

3. Full view (chat app)

Open it with **Full view** under the conversation in the Actions tab, or the expand icon on any job card. It is a maximised window in your current theme, laid out like a chat app:

- **Sidebar:** New chat, your saved **Chats**, and live **Jobs** with status.
- **Centre:** the whole conversation at reading width; answers are full length and selectable.
- **Composer:** follow-up prompt (Enter sends), Continue chip, attachment chips, Attach, drag-and-drop files.

It shares everything with the overlay: a prompt sent in one shows in the other. Closing the window only hides it.

4. Saved chats and resuming

Every chat is saved automatically when an answer finishes: your prompts, the CLI's answers and steps, the CLI/model badge, and the session id needed to resume. Files live in `%LOCALAPPDATA%\AiCockpit\chats\` (one JSON file per chat).

- **Resume:** Full view → Chats → click a chat. The history loads and the Continue chip is set, so your next prompt continues it.
- **Claude chats** resume the real session (`claude --resume`): full context.
- **Antigravity chats**, or switching CLI, carry the earlier turns (about the last 4,000 characters) into the next prompt as context.
- **New chat** starts fresh; **×** deletes a saved chat.

5. Actions, including your own

Action	Use
Ask & learn	General questions and tutoring on any topic. Reads PDFs you attach or a folder you select; linear-algebra results are verified exactly.
Assemble SolidWorks parts	Reads parts in a folder, proposes mates for your review, builds a new .SLDASM.
Design parametric parts	build123d code → STEP → SolidWorks parts, checked against size envelopes.
Organize a folder	Plans moves from your description; you review; every move can be rolled back.
Rough cut for DaVinci Resolve	Probes clips, plans a cut, exports FCPXML + EDL.
Blender car scene	Procedural city scene with a chase camera and a preview render.
Claude Code in a terminal	Opens Claude Code in a folder with your prompt; you drive it.

Native parametric SolidWorks parts

Describe a single part by its dimensions (plates, holes, bosses, ribs, fillets) and run **Design parametric parts** with a Save to folder. The AI builds real SolidWorks features: every size is a global variable (Tools → Equations) and every sketch dimension is linked to them, so the part keeps its design intent. Ask a follow-up such as "change the base to 150 × 80" and it updates the globals, rebuilds, and reports rebuild errors, the bounding box and each hole's distance to the edges. You can also edit the globals yourself in Tools → Equations and press Ctrl+Q. SolidWorks is started automatically if it is not running.

Creating your own action

1. Actions tab → scroll below the tiles → **+ New action**.
2. Give it a **Name**, an optional description, and **Instructions** (what the AI should always do, e.g. "Summarise the attached paper in 10 bullet points, then list open questions").
3. Tick **Needs a folder** if it works on files in a folder selected in Explorer or named in the prompt.
4. **Save action**. It appears as a tile; click **Run** and whatever is in the prompt box is added as input. The trash icon deletes it.

Custom actions are stored in `%LOCALAPPDATA%\AiCockpit\settings.json`. The Cockpit tools they may use are inferred from the instructions and prompt (e.g. mentioning SolidWorks or a PDF).

6. Attachments and folders

A job's folder is chosen in this order: a folder path typed in the prompt, then the **Save to** folder, then the folder or files selected in File Explorer when you pressed the hotkey.

- Select a folder or files in File Explorer *before* pressing the hotkey: they become a context chip (click to include/exclude).
- Attach, drag-and-drop, or Ctrl+V add read-only snapshots. A job never modifies an attachment.
- A folder path typed in the prompt also counts as the job's folder.

7. Routing: Claude Code vs Antigravity

The routing pill cycles **Auto Best** (per-category defaults) → **Quota Saver** (Antigravity first) → **Claude First**. When one subscription is at its usage limit, the job fails over to the other unless "never downgrade" applies. Force a CLI for one prompt with `/claude …` or `/agy …`.

8. Jobs, approvals and the kill switch

- **Jobs tab:** state, CLI, elapsed time, outcome. **Kill** ends one job; the expand icon opens Full view.
- **Approval cards:** risky tool calls (moving files, building assemblies, custom MCP tools) pause for your Approve/Deny.
- **Kill all:** ends every running job immediately.
- **Activity tab:** every tool call with its result, for auditing.

9. Connectors

- **CLIs:** Claude Code and Antigravity, with version and sign-in state. **Sign In** only appears when the CLI is not signed in; it opens a terminal with API-key variables removed so you sign in with your subscription.
- **Desktop apps:** SolidWorks, DaVinci Resolve, Blender. SolidWorks counts as available when installed; if it is not running, the Cockpit starts it when a SolidWorks job needs it.
- **Custom MCP:** see section 12.

Run `cockpit doctor` (in `dist\AiCockpit\`) for a full prerequisite check.

10. Settings and themes

- Themes: Holo HUD, Pro Dark, Clean Light (theme button in the header).
- Hotkey: rebindable; if it conflicts with another app the Cockpit falls back and shows it in the tray tooltip.
- Run on startup: set on the first-run card.
- All settings: `%LOCALAPPDATA%\AiCockpit\settings.json`. Data (jobs, audit, chats, attachments) lives in the same folder.

11. Building a clean release (dist)

From the repository root in PowerShell, with the Cockpit closed (right-click tray icon → Exit):

```
.\scripts\publish-release.ps1 -NoAot -FrameworkDependent
```

- Output: `dist\AiCockpit\` with AiCockpit.exe (overlay), cockpit.exe (CLI host), cockpit-swx-worker.exe, cockpit-mcp-bridge.exe, themes, launchers and this guide. The script finishes with `cockpit doctor`.
- `-NoAot`: builds the bridge as a normal .NET app. Needed on this machine because the .NET 8 Native AOT linker does not work with Visual Studio 18.
- `-FrameworkDependent`: uses the installed .NET 8 Desktop Runtime instead of bundling one per program.

- `-RunTests`: runs the offline test suite first.
- If a build fails with odd file conflicts, delete the `obj` folders under `src\` and run again.

Quick dev run without packaging: `dotnet run --project src\Cockpit.Wpf`. Tests: `.\scripts\run-tests.ps1`.

12. Adding connectors (custom MCP servers)

Any stdio MCP server can be added without code:

1. Connectors tab → **+ Add Custom MCP**.
2. Enter a name, the **full path** to the command (e.g. `C:\Program Files\nodejs\node.exe`) and its arguments.
3. **Check and add**: the Cockpit starts it, pings it (15 s limit) and lists its tools.

- Tools are registered as `ext_<server>_<tool>`. Every call shows an approval card and can never be auto-approved.
- No environment variables are stored (no secrets in `cockpit.mcp.json`). If a server needs a key, wrap it in a script that reads the key itself.
- Servers are saved in `%LOCALAPPDATA%\AiCockpit\cockpit.mcp.json`; you can disable or remove them in the tab.

Built-in connector (code)

For a first-class connector like SolidWorks or Blender: add a tool provider in `src\Cockpit.Core` (see `Tools\ExternalTools.cs` and the tutor/files providers), register its tool names, add a `JobSpecFactory` recipe and an action tile in `DefaultActions()` (`CockpitOverlayViewModel.cs`), and a status probe in `Connectors\ConnectorProbe.cs`.

13. Adding a new CLI

A new AI CLI needs code in these places:

File	Change
`Core\Jobs\ProviderRouter.cs`	Add a value to `AiProvider`, its display name and id, its place in the category defaults, and a prompt prefix in `ParsePrefix`.
`Core\Jobs\CliLaunchBuilder.cs` / `GeminiCli.cs`	A launch builder: headless and interactive command lines, MCP config for the Cockpit bridge, API-key scrubbing.
`Core\Jobs\ClaudeStreamParser.cs`	Parse its output stream into steps, answer text, session id and usage.
`Core\Runtime\CockpitRuntime.cs`	Locate the executable and start headless/interactive jobs with the new builder.
`Core\Connectors\ConnectorProbe.cs`	Version probe, credential file location, sign-in command.
`UI\ConnectorsViewModel.cs`	Add a row to `Clis` and a subscription pill in the overlay.

Then add a test with a recorded output stream under `verify\fixtures\real\` and run the test suite.

14. Troubleshooting

Symptom	Fix
Rebuilt but the exe has no new features	You ran the old `dist` copy. Close the Cockpit and run the publish script (section 11).
"not signed in"	Connectors → Sign In, or run `claude` / `agy` once in a terminal.
SolidWorks job fails with elevation mismatch	Start SolidWorks normally, not "Run as administrator", and don't run the Cockpit elevated.
Resume fails	Claude keeps sessions per working folder; if that folder was removed, start a new chat.
Anything else	Run `cockpit doctor`; the Activity tab shows every tool call and error.